

7) Observation

Ans:-

- The perceptron was trained using the AND gate dataset for 10 epochs.
- During the initial epochs, prediction errors occurred, causing the weights and bias to be updated.
- By Epoch 5, the perceptron correctly classified all input combinations with zero error.
- The final learned parameters were.
 - Weights $[0.2, 0.1]$
 - Bias -0.2
- The trained perceptron successfully learned the AND gate because it is a linearly separable problem.

8) Result

Ans:-

The Perceptron Learning Algorithm was successfully implemented in Python and tested using the AND gate dataset. After training over multiple epochs, the perceptron learned the correct decision boundary by updating its weights and bias. The trained model correctly classified all four input combinations, ^{confirming} that a single-layer perceptron can successfully solve linearly separable binary classification problems such as the AND gate.